1 Summary

Model (Run-name): model1 (test)

Iteration: 26

TAC: 154879.55245396832

Took: 2.4012532234191895s

Total Time: 46.76992845535278s

Epsilons:

- ε_{LMTD}: 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area}: 1e-06
- ε_{Beta}: 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

Gamsworld Solution: `http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html`

2 Results

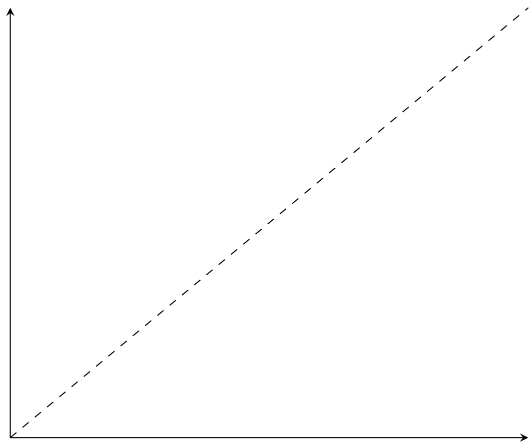
Variable	Value
$z_{1,1,1}$	1.0
$z_{1,2,2}$	1.0
$z_{2,1,2}$	1.0
$z_{cu,1}$	1.0
$z_{cu,2}$	1.0
$z_{hu,1}$	1.0
$A^{\beta}_{1,1,1}$	72.36161285231599
$A^{\beta}_{1,2,2}$	72.87506706799363
$A^{\beta}_{2,1,2}$	127.07690270456857
$A^{\beta}_{cu,1}$	4.0691182452509045
$A^{\beta}_{cu,2}$	38.224970554765775
$A^{\beta}_{hu,1}$	13.498843422507207
$A_{1,1,1}$	72.36161285231599
$A_{1,2,2}$	72.87506706799363
$A_{2,1,2}$	127.07690270456857
$A_{cu,1}$	4.0691182452509045
$A_{cu,2}$	38.224970554765775
$A_{hu,1}$	13.498843422507207
$dt_{1,1,1}$	33.62356156270736
$dt_{1,1,2}$	10.0
$dt_{1,2,2}$	79.12931531187803
$dt_{1,2,3}$	34.12931531187803
$dt_{2,1,2}$	20.87068468812197
$dt_{2,1,3}$	60.653013516091505
$dt_{cu,1}$	64.12931531187803
$dt_{cu,2}$	150.6530135160915
$dt_{hu,1}$	63.62356156270736
$f^{H}_{1,1,1}$	2.5310958817186417
$f^{H}_{1,2,2}$	10.0
$f^{H}_{2,1,2}$	10.849726043991682
$f^{C}_{1,1,1}$	15.0
$f^{C}_{1,2,2}$	13.0
$f^{C}_{2,1,2}$	10.522725094359455

Variable	Value
$RecLMTD_{1,1,1}$	0.051331491549091945
$RecLMTD_{1,2,2}$	0.018685914632818892
$RecLMTD_{2,1,2}$	0.026812457899271867
$RecLMTD_{cu,1}$	0.01492030929070222
$RecLMTD_{cu,2}$	0.009502863867464236
$RecLMTD_{hu,1}$	0.022358875540792994
$q_{1,1,1}$	708.7068468812197
$q_{1,2,2}$	1950.0
$q_{2,1,2}$	2386.93972967817
$q_{cu,1}$	141.2931531187803
$q_{cu,2}$	2013.06027032183
$q_{hu,1}$	504.3534234406104
$t^{(H)}_{1,1}$	650.0
$t^{(H)}_{1,2}$	579.129315311878
$t^{(H)}_{1,3}$	384.12931531187803
$t^{(H)}_{2,1}$	590.0
$t^{(H)}_{2,2}$	590.0
$t^{(H)}_{2,3}$	470.6530135160915
$t^{(C)}_{1,1}$	616.3764384372927
$t^{(C)}_{1,2}$	569.129315311878
$t^{(C)}_{1,3}$	410.0
$t^{(C)}_{2,1}$	500.0
$t^{(C)}_{2,2}$	500.0
$t^{(C)}_{2,3}$	350.0
$t^{H}_{1,1,1}$	370.0
$t^{H}_{1,2,2}$	384.1293153118955
$t^{H}_{2,1,2}$	370.0
$t^{C}_{1,1,1}$	616.3764384372927
$t^{C}_{1,2,2}$	500.0
$t^{C}_{2,1,2}$	633.3645302695838
$b^{H,\text{in}}_{1,1,1}$	1645.2123231171172
$b^{H,\text{in}}_{1,2,2}$	5791.2931531187805
$b^{H,\text{in}}_{2,1,2}$	6401.338365955092

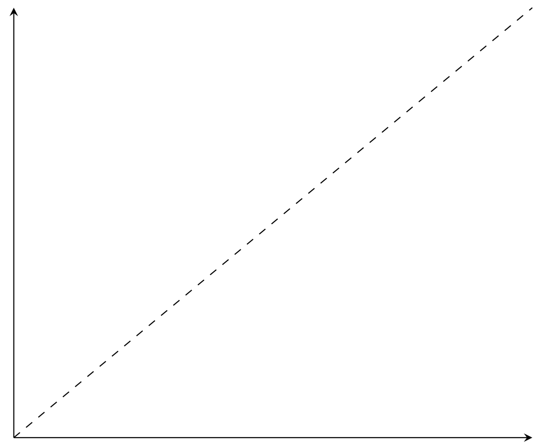
Variable	Value
$b^{H,\text{out}}_{1,1,1}$	936.5054762358975
$b^{H,\text{out}}_{1,2,2}$	3841.2931531187805
$b^{H,\text{out}}_{2,1,2}$	4014.3986362769224
$b^{C,\text{in}}_{1,1,1}$	8536.93972967817
$b^{C,\text{in}}_{1,2,2}$	4550.0
$b^{C,\text{in}}_{2,1,2}$	4314.317288687376
$b^{C,\text{out}}_{1,1,1}$	9245.64657655939
$b^{C,\text{out}}_{1,2,2}$	6500.0
$b^{C,\text{out}}_{2,1,2}$	6701.257018365547

3 *RecLMTD* Tangent Points

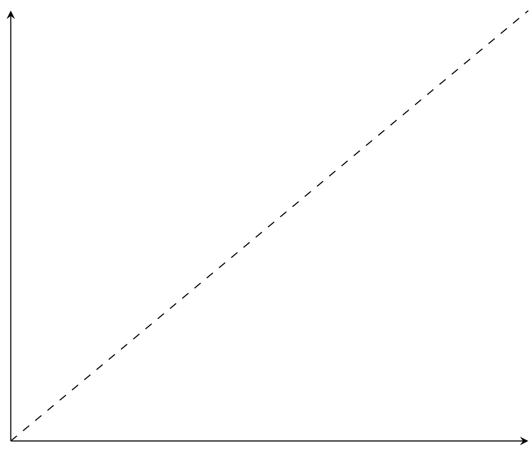
- $(1, 1, 1)$
  - Value: 0.051331
  - Error: 0.000000394288265
  - Relative error: 0.000007681156827
- $(1, 2, 2)$
  - Value: 0.018686
  - Error: 0.000001345708249
  - Relative error: 0.000072012067299
- $(2, 1, 2)$
  - Value: 0.026812
  - Error: 0.000004066541050
  - Relative error: 0.000151643105681
- $cu, 1$
  - Value: 0.014920
  - Error: 0.000000206826939
  - Relative error: 0.000013861915881
- $cu, 2$
  - Value: 0.009503
  - Error: 0.000000612733565
  - Relative error: 0.000064474674951
- $hu, 1$
  - Value: 0.022359
  - Error: 0.000000043323264
  - Relative error: 0.000001937627876



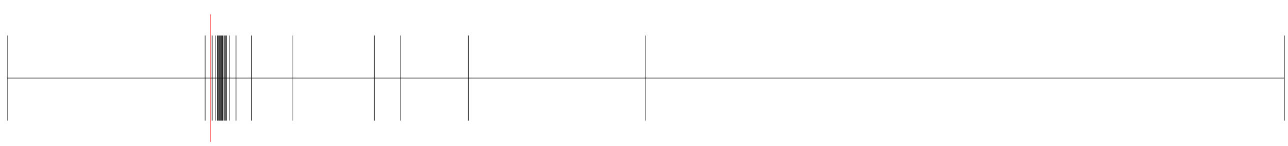
(a) *RecLMTD* - $(1, 1, 1)$



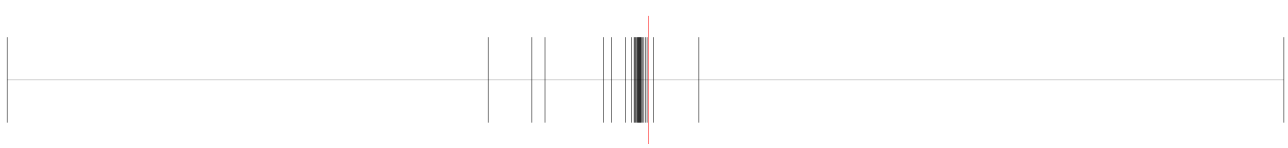
(b) *RecLMTD* - $(1, 2, 2)$



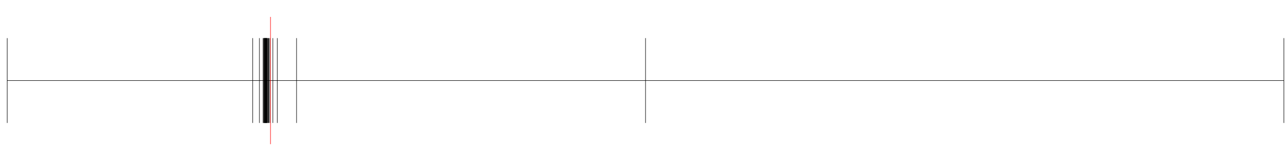
(c) *RecLMTD* - $(2, 1, 2)$



(a) *RecLMTD* - $cu, 1$



(b) *RecLMTD* - $cu, 2$



(a) *RecLMTD* - $hu, 1$

4 Balancing Breakpoints

4.1 $b^{\boldsymbol{H},\mathbf{in}}$

- $(1, 1, 1)$
  - Value: 1645.2123231171172
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 5791.2931531187805
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 6401.338365955092
  - Absolute error: 0.0
  - Relative error: 0.0

4.2 $b^{\boldsymbol{H},\mathbf{out}}$

- $(1, 1, 1)$
  - Value: 936.5054762358975
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 3841.2931531187805
  - Absolute error: 1.7416823538951576e-10
  - Relative error: -4.5341042312300245e-14
- $(2, 1, 2)$
  - Value: 4014.3986362769224
  - Absolute error: 0.0
  - Relative error: 0.0

4.3 $b^{\boldsymbol{C},\mathbf{in}}$

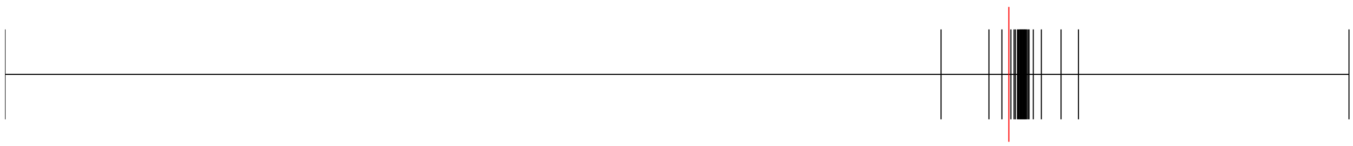
- $(1, 1, 1)$
  - Value: 8536.93972967817
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 4550.0
  - Absolute error: 0.0

  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 4314.317288687376
  - Absolute error: 0.0
  - Relative error: 0.0

4.4 $b^{\mathbf{C,out}}$

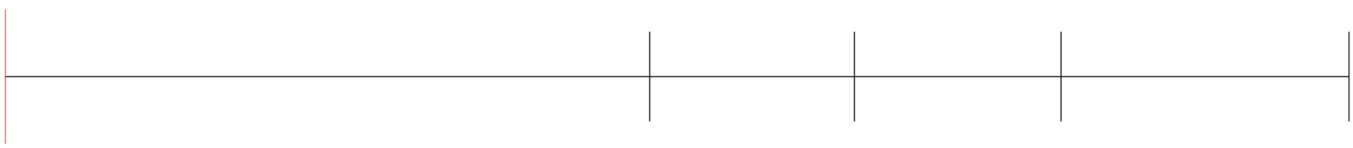
- $(1, 1, 1)$
  - Value: 9245.64657655939
  - Absolute error: 0.0
  - Relative error: 0.0
- $(1, 2, 2)$
  - Value: 6500.0
  - Absolute error: 0.0
  - Relative error: 0.0
- $(2, 1, 2)$
  - Value: 6701.257018365547
  - Absolute error: 36.53618182060836
  - Relative error: 0.005482027337179378

(a) $t^{(H)}$ - $(1,1)$



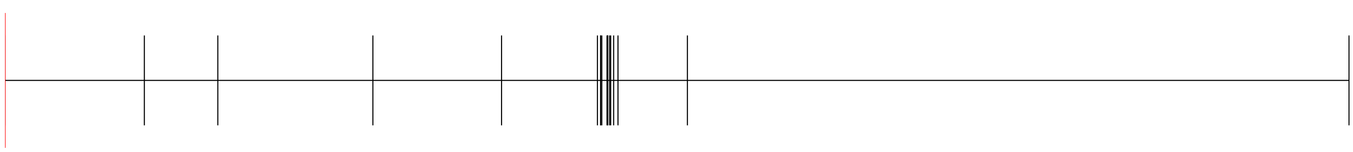
(a) $t^{(H)}$ - $(1,2)$

(a) $t^{(H)}$ - $(2,2)$



(a) $t^{(H)}$ - $(1,1,1)$

(a) $t^{(H)}$ - $(1,2,2)$



(a) $t^{(H)}$ - $(2,1,2)$

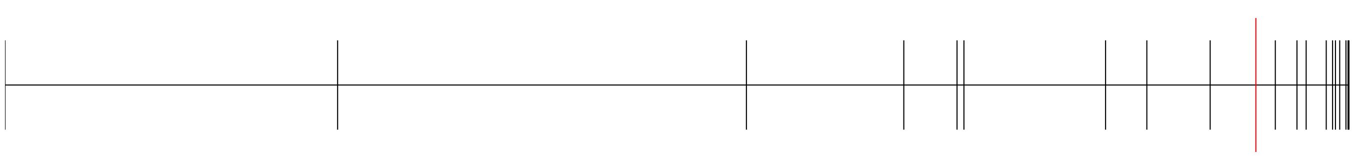
(a) $t^{(C)}$ - $(1,2)$

(a) $t^{(C)}$ - $(2,3)$

(a) $t^{(C)}$ - $(1, 3)$

(a) $t^{(C)}$ - $(1, 1, 1)$

(a) $t^{(C)}$ - $(1, 2, 2)$



(a) $t^{(C)}$ - $(2, 1, 2)$

5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
  - Value: 72.361613
  - Error: 0.396346190617876
  - Relative error: 0.005447461636245
- $(1, 2, 2)$
  - Value: 72.875067
  - Error: 0.000000000000057
  - Relative error: 0.000000000000001
- $(2, 1, 2)$
  - Value: 127.076903
  - Error: 0.922539315622032
  - Relative error: 0.007207369821788

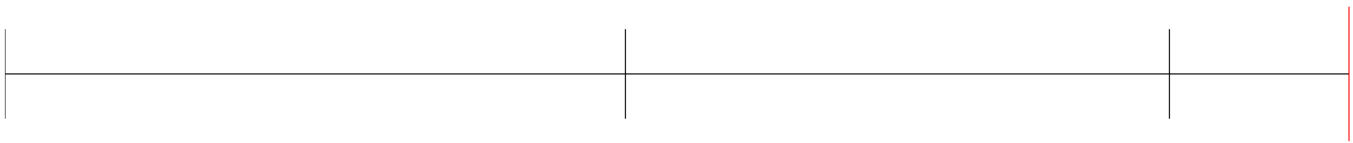
5.2 A_{CU}

- $cu, 1$
  - Value: 4.069118
  - Error: 0.147156845130594
  - Relative error: 0.034902097699057
- $cu, 2$
  - Value: 38.224971
  - Error: 0.034704856972439
  - Relative error: 0.000907087072720

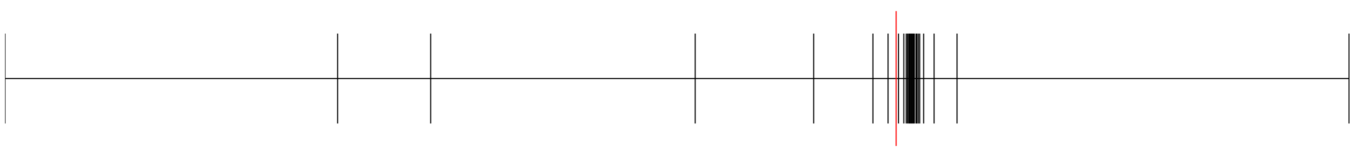
5.3 A_{HU}

- $hu, 1$
  - Value: 13.498843
  - Error: 0.033287085430564
  - Relative error: 0.002459855483291

(a) A^β - $(1, 1, 1)$

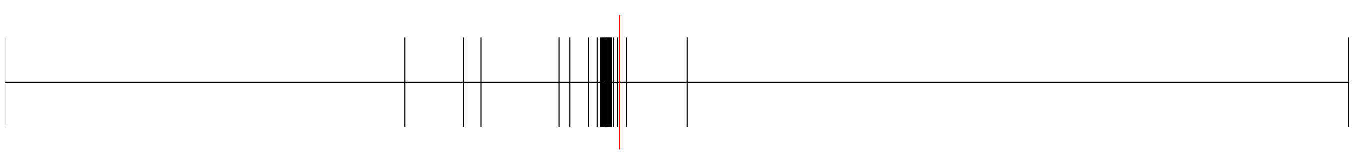


(a) A^β - $(1, 2, 2)$



(a) A^β - $(2, 1, 2)$

(a) A^β - $cu, 1$



(a) A^β - $cu, 2$

(a) A^β - $hu, 1$

6 A^β Breakpoints

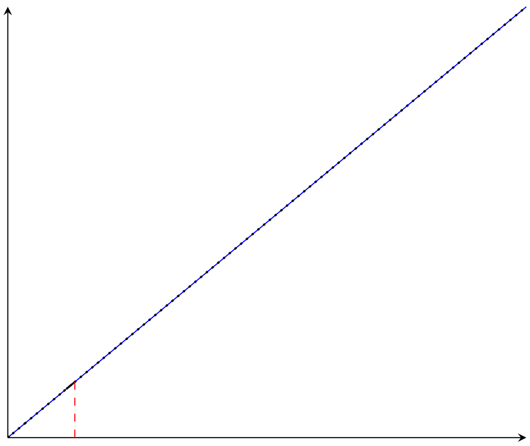
- $(1,1,1)$
  - q: 72.361613
  - $\hat{q}^\beta$: 72.361613
  - q^β: 72.361613
  - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 63.3468696286352, 65.40595072874316, 67.27362048222933, 69.14107665413653, 69.42777867315647, 69.89902342903967, 70.11420238741596, 70.22290217272871, 70.30181758862284, 70.46247894104941, 70.50425945122109, 70.60541450524272, 70.65581083979713, 70.83181816751056, 70.95870595008942, 71.00935653192329, 71.18527282143235, 71.22933662920877, 71.2616536557244, 71.33750386361461, 71.6637831223933, 71.88123217857378, 560.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(1,2,2)$
  - q: 72.875067
  - $\hat{q}^\beta$: 72.875067
  - q^β: 72.875067
  - Breakpoints: [0, 6.000977230027827, 11.262728503004022, 39.36524453694061, 64.01825678947372, 66.17325582523148, 67.27673099173637, 67.60071888621576, 67.91344539650363, 68.10969007738701, 68.48329505760876, 68.66676327863097, 68.85281320938876, 69.0314457108091, 69.22655548799992, 69.34502840170047, 69.41355730879985, 69.60484346150977, 70.08419437796795, 70.37078617087542, 70.94081287188642, 72.25348351595485, 74.6496383881616, 84.1468297226002, 390.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $(2,1,2)$
  - q: 127.076903
  - $\hat{q}^\beta$: 127.076903
  - q^β: 127.076903
  - Breakpoints: [0, 9.898970403313276, 28.333333333333332, 42.69900440382105, 48.67452649027746, 79.83702775273511, 87.16107999267206, 118.88765652962957, 128.52021081477858, 131.69760457629607, 132.12228195956155, 135.81963163113545, 137.02923097052596, 138.86359281280824, 139.47623233772285, 140.31685261310218, 140.7008934199032, 141.79783607898793, 142.4338700260237, 143.30274961189664, 144.67005099128556, 144.70843093515643, 147.02575214945074, 147.07009017600456, 147.48374761404892, 149.84689443310864, 720.0]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000
- $cu, 1$
  - q: 4.069118
  - $\hat{q}^\beta$: 4.069118
  - q^β: 4.069118
  - Breakpoints: [0, 2.5450346292088537, 2.9819181007907165, 3.1809419553840232, 3.9948823402911255, 4.269280092716329, 4.432280503220532, 4.613047202566896, 4.707108999269609, 4.813467676447643, 4.825601156120121, 4.849719301830286, 4.888121952381175, 4.962276317638335, 4.974201802143407, 4.983938626310074, 4.993068227518925,

5.0359974771272995, 5.0418124100090695, 5.081618324442152, 5.195781532076625, 7.702824927089129, 9.28003023700696, 13.56934656224007, 14.535545040173215, 181.61828057849593]

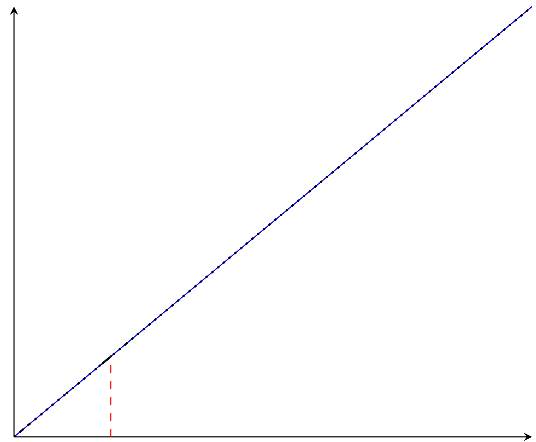
- Error: 0.000000000000000
- Relative error: 0.000000000000000

- $cu, 2$
  - q: 38.224971
  - $\hat{q}^\beta$: 38.224971
  - q^β: 38.224971
  - Breakpoints: [0, 18.023935870161154, 21.04660840930274, 24.97364426355679, 30.189424467922265, 34.23465128608717, 36.670233878040335, 37.21616748032837, 37.443498367229196, 37.47875108132638, 37.52720514437563, 37.554741031910254, 37.612770977560544, 37.65435035996605, 37.6551299822246, 37.682669717644174, 37.71812813120871, 37.76995088018704, 37.78573239590169, 37.821651654621526, 37.88148462263016, 37.9018730666985, 37.9454585315849, 38.10872593446926, 38.21652191748154, 38.88483572219945, 285.4001551947793]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000

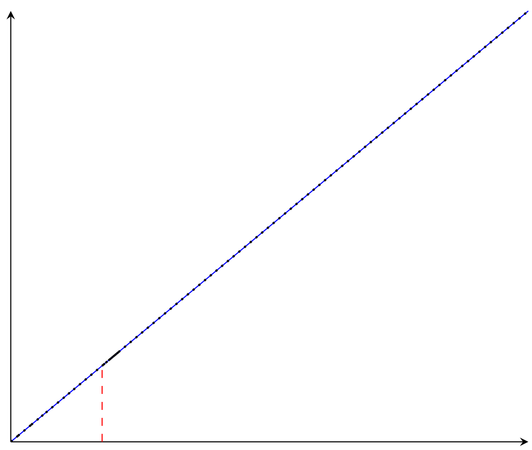
- $hu, 1$
  - q: 13.498843
  - $\hat{q}^\beta$: 13.498843
  - q^β: 13.498843
  - Breakpoints: [0, 4.943755299006499, 12.544341108707222, 12.595096067947068, 12.917705220652834, 13.104034300095861, 13.134779411775817, 13.161020122653593, 13.189852638760229, 13.192252946941991, 13.211215176722128, 13.213749911872785, 13.218222920629291, 13.22717920580809, 13.238585544475612, 13.244985069124171, 13.266498143734903, 13.269097009439808, 13.294574825457861, 13.30871858348944, 13.32330648750218, 13.32663103908035, 13.346967780023082, 13.439500729482463, 13.468285806082731, 14.09178614751222, 237.3002543523117]
  - Error: 0.000000000000000
  - Relative error: 0.000000000000000



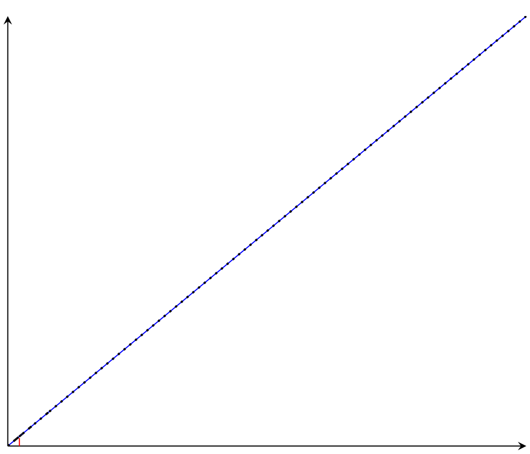
(a) q^β - $(1,1,1)$



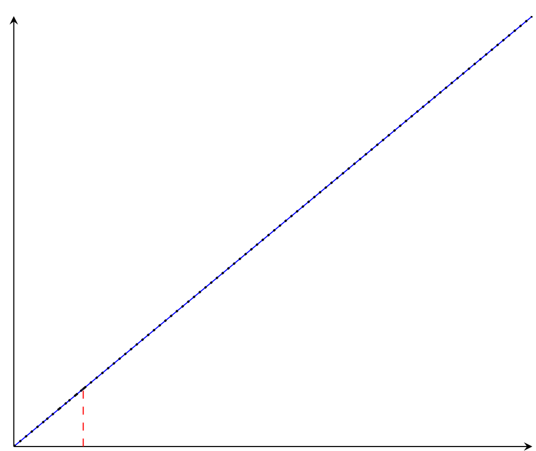
(b) q^β - $(1,2,2)$



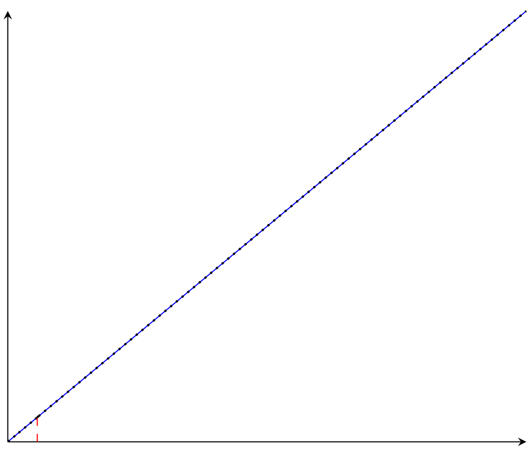
(c) q^β - $(2,1,2)$



(a) q^β - $cu, 1$



(b) q^β - $cu, 2$



(a) q^β - $hu, 1$

7 Other variables

Variable	Value
$z_{1,1,2}$	-0.0
$z_{1,2,1}$	-0.0
$z_{2,1,1}$	-0.0
$z_{2,2,1}$	-0.0
$z_{2,2,2}$	-0.0
$z_{hu,2}$	-0.0
$A^{\beta}_{1,1,2}$	0.0
$A^{\beta}_{1,2,1}$	0.0
$A^{\beta}_{2,1,1}$	0.0
$A^{\beta}_{2,2,1}$	0.0
$A^{\beta}_{2,2,2}$	0.0
$A^{\beta}_{hu,2}$	0.0
$A_{1,1,2}$	0.0
$A_{1,2,1}$	0.0
$A_{2,1,1}$	0.0
$A_{2,2,1}$	0.0
$A_{2,2,2}$	0.0
$A_{hu,2}$	0.0
$dt_{1,1,3}$	240.0
$dt_{1,2,1}$	150.0
$dt_{2,1,1}$	24.166666666666668
$dt_{2,2,1}$	90.0
$dt_{2,2,2}$	10.0
$dt_{2,2,3}$	240.0
$dt_{hu,2}$	180.0
$f^{H}_{1,1,2}$	0.0
$f^{H}_{1,2,1}$	7.468904118281358
$f^{H}_{2,1,1}$	20.0
$f^{H}_{2,2,1}$	0.0
$f^{H}_{2,2,2}$	9.150273956008318
$f^{C}_{1,1,2}$	4.477274905640545
$f^{C}_{1,2,1}$	13.0
$f^{C}_{2,1,1}$	0.0

Variable	Value
$f^{C}_{2,2,1}$	0.0
$f^{C}_{2,2,2}$	0.0
$RecLMTD_{1,1,2}$	0.1
$RecLMTD_{1,2,1}$	0.008134665654279333
$RecLMTD_{2,1,1}$	0.1
$RecLMTD_{2,2,1}$	0.020111728838038176
$RecLMTD_{2,2,2}$	0.013817625349338898
$RecLMTD_{hu,2}$	0.005549329826813255
$q_{1,1,2}$	0.0
$q_{1,2,1}$	0.0
$q_{2,1,1}$	0.0
$q_{2,2,1}$	0.0
$q_{2,2,2}$	0.0
$q_{hu,2}$	0.0
$t^{H}_{1,1,2}$	418.8196134269507
$t^{H}_{1,2,1}$	650.0
$t^{H}_{2,1,1}$	590.0
$t^{H}_{2,2,1}$	590.0
$t^{H}_{2,2,2}$	583.7639441750986
$t^{C}_{1,1,2}$	410.0
$t^{C}_{1,2,1}$	500.0
$t^{C}_{2,1,1}$	410.0
$t^{C}_{2,2,1}$	350.0
$t^{C}_{2,2,2}$	440.01120358475225
$b^{H,\text{in}}_{1,1,2}$	0.0
$b^{H,\text{in}}_{1,2,1}$	4854.787676882883
$b^{H,\text{in}}_{2,1,1}$	11800.0
$b^{H,\text{in}}_{2,2,1}$	0.0
$b^{H,\text{in}}_{2,2,2}$	5398.661634044907
$b^{H,\text{out}}_{1,1,2}$	0.0
$b^{H,\text{out}}_{1,2,1}$	4854.787676882883
$b^{H,\text{out}}_{2,1,1}$	11800.0
$b^{H,\text{out}}_{2,2,1}$	0.0

Variable	Value
$b^{H,\text{out}}_{2,2,2}$	5398.661634044907
$b^{C,\text{in}}_{1,1,2}$	1835.6827113126235
$b^{C,\text{in}}_{1,2,1}$	6500.0
$b^{C,\text{in}}_{2,1,1}$	0.0
$b^{C,\text{in}}_{2,2,1}$	0.0
$b^{C,\text{in}}_{2,2,2}$	0.0
$b^{C,\text{out}}_{1,1,2}$	1835.6827113126228
$b^{C,\text{out}}_{1,2,1}$	6500.0
$b^{C,\text{out}}_{2,1,1}$	0.0
$b^{C,\text{out}}_{2,2,1}$	0.0
$b^{C,\text{out}}_{2,2,2}$	0.0